William Thompson

Plaskett Fellow · National Research Council Canada - Herzberg Astronomy and Astrophysics
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Research Interests

Exoplanet Detection, Absolute Astrometry, Exoplanet Direct Imaging, Doppler Radial Velocity, Astronomical Instrumentation

Education

PhD in Astronomy 2018-2023

University of Victoria · GPA: 9.0/9.0 · Supervisors: C. Marois and F. Herwig
Thesis: "Lights in Motion: Observing Nearby Planets with Imaging, Wavefront Sensing, Orbital Detection, and Spectroscopy"

Bachelor of Science (Honours) in Physics with Distinction 2018

Queen's University

Bachelor of Computing with Distinction 2016

Queen's University

Academic Employment

Assistant Professor from July 2027

University of Victoria, Department of Physics and Astronomy

Plaskett Fellow (Assistant Research Officer) 2025-present

National Research Council Canada - Herzberg Astronomy and Astrophysics

Herzberg Instrument Science Fellow (Assistant Research Officer) 2023-2025

National Research Council Canada - Herzberg Astronomy and Astrophysics

Awards and Honors

Competitive Grants and Scholarships ($272,780 total)

2026	$59,400	CSA JWST Cycle 4 AO	(Co-PI of grant, Co-PI of proposal)
2026	$34,980	CSA JWST Cycle 4 AO	(Co-PI of grant, Co-I of proposal)
2025	$19,000	NRC HAA Strategic R&D Proposal	(PI)
2024	$48,400	CSA JWST Cycle 3 AO	(Co-PI of grant, Co-I of proposal)
2023	$5,000	President's Research Scholarship (University of Victoria Faculty of Graduate Studies)	
2022	$63,000	PGS-D Scholarship (NSERC)	
2022	$5,000	President's Research Scholarship (University of Victoria Faculty of Graduate Studies)	
2021	$3,500	Alexander and Helen Stafford MacCarthy Muir Graduate Scholarship	
2019	$12,000	New Technologies for Canadian Observatories (NSERC)	
2018	$17,500	CGS-M Scholarship (NSERC CREATE)	
2018	$5,000	President's Research Scholarship (University of Victoria Faculty of Graduate Studies)	

Honors

- **2025**: Plaskett Medal for most outstanding doctoral thesis in Canada
- **2022**: Best Poster at Canadian Astronomical Society AGM
- **2022**: Astro Plot of the Week (online poll)
- **2021**: Runner up best student presentation, NRC Herzberg Jamboree
- **2019**: Nominated for Vanier Fellowship to national level by the University of Victoria
- **2019**: Best Poster at Canadian Astronomical Society AGM
- **2016**: International semi-finalist (innovation) Microsoft Imagine Cup

Publications

Journal Articles: 30 (7 first-author) · **Conference Proceedings: 20** · **White Papers: 1**

Journal Articles

2026

1. **Thompson W.**, Blakely D., Xuan J.W., Blouin S., Zhang J., Johnstone D., Ruffio J.-B., Nielsen E., Speedie J., Bowler B., Bouchard-Côté A., Franson K., Blunt S., Roberson W., Cloutier R., Fogal A., Hessel K., Marois C., Rochon A.. "Detecting and Characterizing Companions with a Calibrated Gaia DR2, DR3, and Hipparcos Catalog (G23H)." *The Astronomical Journal 172, 53*. 2026. [1 citations]

2. Sanghi A., **Thompson W.**, Mang J., Xuan J., Mawet D., Ruffio J.-B., Zhang Y., et al.. "Worlds Next Door. IV. Mapping the Late Stages of Giant Planet Evolution with a Precise Dynamical Mass and Luminosity for ε Ind Ab." *arXiv preprint*. 2026. *(preprint)*

3. Brinjikji M., Smith A.J., Patience J., Nielsen E., Ware A., Sanchez J.A., et al. (incl. **Thompson W.**). "The Companions to B and A Stars Snapshot (C-BASS) Survey: I. Discovery of a Young Brown Dwarf Companion to HIP 17453." *arXiv preprint*. 2026. *(preprint)*

4. Van Zandt J., Bowler B., Petigura E., Franson K., Xuan J., Morgan M., et al. (incl. **Thompson W.**). "Gaia Exoplanet Orbits, Demographics, and Evolution Survey (GEODES): Characteristics of Three Long-Period Companions Accelerating their Host Stars." *arXiv preprint*. 2026. *(preprint)*

5. Blakely D., **Thompson W.**, Johnstone D., Speedie J., Xuan J.W., Blouin S., Zhang J., Ruffio J.-B., Nielsen E., Bowler B.P., Franson K., Roberson W., Cloutier R., Fogal A., Hessel K., Marois C., Rochon A.. "Dynamical Mass Constraints on Transition Disk Perturbers with the G23H Catalog." *arXiv preprint*. 2026. *(preprint)* [1 citations]

6. Sanghi A., Mang J., Llop-Sayson J., Mamajek E.E., **Thompson W.**, Sur A., et al.. "Worlds Next Door. III. Indirect Evidence for Enhanced Atmospheric Metallicity and/or the Presence of Water Clouds in the Nearest Jupiter-analog ε Eri b." *The Astronomical Journal 171, 225*. 2026. [2 citations]

7. Currie T., Li Y., El Morsy M., Lacy B., Vincent M., Tobin T.L., Kuzuhara M., et al. (incl. **Thompson W.**). "SCExAO/CHARIS and Gaia Direct Imaging and Astrometric Discovery of a Superjovian Planet 3-4 λ/D from the Accelerating Star HIP 54515." *The Astronomical Journal 171, 5*. 2026. [6 citations]

2025

8. Bowens-Rubin R., Salama M., Nguyen J.S., **Thompson W.**, Hinz P.. "An on-sky investigation into factors limiting the performance of Keck-NIRC2 for conducting infrared high-contrast imaging." *arXiv preprint*. 2025. *(preprint)*

9. **Thompson W.**, Nielsen E.L., Ruffio J.-B., Blunt S., Marois C.. "Revised Mass and Orbit of ε Eridani b: A 1 Jupiter-Mass Planet on a Near-Circular Orbit." *The Astronomical Journal 170, 301*. 2025. [3 citations]

10. **Thompson W.**, Blakely D., Xuan J.W., Bouchard-Côté A., Bourdarot G., Biron-Lattes M., Campbell T., Eisenhauer F., Henning T., Janson M., Johnstone D., Kammerer J., Konopacky Q., Lacour S., Marois C., Mawet D., Mérand A., Nguyen J.S., Nielsen E., Rickman E., Ruffio J.-B., Surjanovic N., Wang J.J., Winterhalder T.. "On the Orbit of the Binary Brown Dwarf Companion GL229 Ba and Bb." *The Astronomical Journal 169, 193*. 2025. [2 citations]

11. Nguyen J.S., Konopacky Q.M., **Thompson W**., Popenoe N., Macintosh B.. "Ground-Based Mid-IR Direct Imaging: The Origin of the Thermal Background on the Keck II Telescope and Correcting Instrumental Systematics." *Publications of the Astronomical Society of the Pacific 137, 95002*. 2025.

12. Roberts J., **Thompson W**., Wang J.J., Blunt S., Balmer W.O., Bourdarot G., et al.. "New Orbital Constraints for YSES 1 b and HR 2562 B from High-Precision Astrometry and Planetary Radial Velocities." *The Astronomical Journal 170, 273*. 2025.

13. Sagynbayeva S., Abbas A., Kane S.R., Nielsen E.L., **Thompson W**., Blunt S., Rice M., et al.. "Requirements for Joint Orbital Characterization of Cold Giants and Habitable Worlds with Habitable Worlds Observatory." *The Astronomical Journal 170, 208*. 2025.

14. Llop-Sayson J., Beichman C., Bryden G., Ygouf M., Gaspar A., **Thompson W**., Sanghi A., Mawet D., et al.. "Searching for Planets Orbiting ε Eridani with JWST/NIRCam." *The Astronomical Journal 170, 229*. 2025. [2 citations]

15. Peck A.E., Nielsen E.L., De Rosa R.J., **Thompson W**., Macintosh B., Roberson W., Smith A.J.R.W., Klusmeyer J., Abbas A., Jackiewicz J., Holtzman J., Gallamore H., Brinjikji M., Patience J., Nguyen J.S., Madurowicz A., Savransky D.. "Apache Point Observatory Follow-up of ACcelerating Candidate ExopLanet Host Stars (APO ACCELS): Ages for 166 Accelerating Stars in the Northern Hemisphere." *The Astronomical Journal 170, 122*. 2025. [1 citations]

16. Hammond I., Christiaens V., Price D.J., Blakely D., Trevascus D., Bonse M.J., Cantalloube F., Marleau G.-D., Pinte C., Juillard S., Samland M., **Thompson W**., Wallace A.. "Keplerian motion of a compact source orbiting the inner disc of PDS 70: a third protoplanet in resonance with b and c?." *Monthly Notices of the Royal Astronomical Society 539, 1613*. 2025. [6 citations]

17. El Morsy M., Currie T., Lacy B., Tobin T.L., An Q., Li Y., Gu Z., Kuzuhara M., et al. (incl. **Thompson W**.). "OASIS Survey Direct Imaging and Astrometric Discovery of HIP 71618 B: A Substellar Companion Suitable for the Roman Coronagraph Technology Demonstration." *The Astrophysical Journal Letters 995, L4*. 2025. [3 citations]

18. Blakely D., Johnstone D., Stolker T., Benisty M., Kammerer J., Norfolk B.J., **Thompson W**., Berger J.-P.. "A Tentative Detection of a Point Source in the Disk Gap of HD 100546 with VLT/SPHERE-IRDIS Sparse Aperture Masking Interferometry." *The Astronomical Journal 169, 152*. 2025. [2 citations]

19. Blakely D., Johnstone D., Cugno G., Sivaramakrishnan A., Tuthill P., Dong R., Pope B.J.S., et al. (incl. **Thompson W**.). "The James Webb Interferometer: Space-based Interferometric Detections of PDS 70 b and c at 4.8 μm." *The Astronomical Journal 169, 137*. 2025. [17 citations]

2024

20. Xuan J.W., Mérand A.*, **Thompson W**.*, Zhang Y., Lacour S., Blakely D., Mawet D., Oppenheimer R., Kammerer J., Batygin K., et al.. "The cool brown dwarf Gliese 229B is a close binary." *Nature 634, 1070-1074*. 2024. *Both authors contributed equally to this work [24 citations]

21. Mao H., Woodward P., Herwig F., Denissenkov P.A., Blouin S., **Thompson W**., McDermott B.. "3D Hydrodynamic Simulations of Massive Main-sequence Stars. III. The Effect of Radiation Pressure and Diffusion Leading to a 1D Equilibrium Model." *The Astrophysical Journal 975, 271*. 2024. [11 citations]

22. **Thompson W**., Herwig F., Woodward P.R., Mao H., Denissenkov P., Bowman D.M., Blouin S.. "3D hydrodynamic simulations of massive main-sequence stars II. Convective excitation and spectra of internal gravity waves." *Monthly Notices of the Royal Astronomical Society 531, 1316-1337*. 2024. [25 citations]

23. Do Ó C.R., Sappey B., Konopacky Q.M., Ruffio J.-B., O'Neil K.K., Do T., Martinez G., Barman T.S., Nguyen J.S., Xuan J.W., Theissen C.A., Blunt S., **Thompson W**., et al.. "Orbital and Atmospheric Characterization of the 1RXS J034231.8+121622 System using High-resolution Spectroscopy Confirms that the Companion is a Low-mass Star." *The Astronomical Journal 167, 278*. 2024. [5 citations]

2023

24. Currie T., Brandt G.M., Brandt T.D., Lacy B., Burrows A., Guyon O., Tamura M., et al. (incl. **Thompson W**.). "Direct imaging and astrometric detection of a gas giant planet orbiting an accelerating star." *Science 380, 198-203*. 2023. [76

citations]

25. Herwig F., Woodward P.R., Mao H., **Thompson W**.R., Denissenkov P., Lau J., Blouin S., Andrassy R., Paul A.. "3D hydrodynamic simulations of massive main-sequence stars I. Dynamics and mixing of convection and internal gravity waves." *Monthly Notices of the Royal Astronomical Society 525, 1601-1629*. 2023. [37 citations]
26. Blouin S., Mao H., Herwig F., Denissenkov P., Woodward P.R., **Thompson W**.R.. "3D hydrodynamics simulations of internal gravity waves in red giant branch stars." *Monthly Notices of the Royal Astronomical Society 522, 1706-1725*. 2023. [19 citations]
27. **Thompson W**., Lawrence J., Blakely D., Marois C., Wang J., Giordano M., Brandt T., Johnstone D., Ruffio J.-B., Ammons S.M., Crotts K.A., Do Ó C.R., Gonzales E.C., Rice M.. "Octofitter: Fast, Flexible, and Accurate Orbit Modeling to Detect Exoplanets." *The Astronomical Journal 166, 164*. 2023. [17 citations]
28. Agrawal S., Ruffio J.-B., Konopacky Q.M., Macintosh B., Mawet D., Nielsen E.L., Hoch K.K.W., Liu M.C., Barman T.S., **Thompson W**., Greenbaum A.Z., Marois C., Patience J.. "Detecting Exoplanets Closer to Stars with Moderate Spectral Resolution Integral-field Spectroscopy." *The Astronomical Journal 166, 15*. 2023. [12 citations]
29. **Thompson W**., Marois C., Do Ó C.R., Konopacky Q., Ruffio J.-B., Wang J., Skemer A.J., De Rosa R.J., Macintosh B.. "Deep Orbital Search for Additional Planets in the HR 8799 System." *The Astronomical Journal 165, 29*. 2023. [13 citations]

2021

30. **Thompson W**., Marois C.. "Improved Contrast in Images of Exoplanets Using Direct Signal-to-noise Ratio Optimization." *The Astronomical Journal 161(5), 236*. 2021. [16 citations]

Conference Proceedings

1. **Thompson W**.R., Johnson A., Lardière O., Marois C., Gamroth D., Mann C., et al.. "SPIDERS: a pathfinder 4th generation planet imager." *Adaptive Optics Systems IX, Proceedings of the SPIE 13097, 130974K*. 2024. [3 citations]
2. Fitzsimmons J., Lardière O., Marois C., **Thompson W**., Johnson A.. "NRC integrated modeling: experimental validation with SPIDERS." *Modeling, Systems Engineering, and Project Management for Astronomy XI 13099*. 2024. [1 citations]
3. **Thompson W**., Gamroth D., Marois C., Lardière O.. "Real-time adaptive optics control with a high level programming language." *Proceedings of the SPIE*. 2024.
4. **Thompson W**., Giordano M.. "AstroImages.jl." *The Proceedings of the JuliaCon Conferences 6, 57*. 2024.
5. Marois C., Lardière O., Fitzsimmons J., et al. (incl. **Thompson W**.). "CAL2: project update of the NRC Canada facility-class focal plane wavefront sensor for GPI 2.0." *Proceedings of the SPIE*. 2024.
6. Johnson A.B., Marois C., **Thompson W**., et al.. "An optical feasibility study for STARLITE." *Proceedings of the SPIE*. 2024.
7. Mann C., Marois C., **Thompson W**., Lardière O., Véran J.-P.. "Coherent differential imaging: squeezing additional imaging contrast behind the self-coherent camera on SPIDERS." *Proceedings of the SPIE*. 2024.
8. **Thompson W**., Johnson A., Marois C., et al.. "High-Spectral Resolution Dark Holes: Concept, Results, and Promise." *Proceedings of Adaptive Optics for Extremely Large Telescopes 7*. 2023.
9. Lardière O., **Thompson W**., Johnson A., Marois C., Fitzsimmons J., Hardy T., et al.. "Design, integration & tests of SPIDERS: a Subaru Pathfinder Instrument for Detecting Exoplanets & Retrieving Spectra." *Proceedings of Adaptive Optics for Extremely Large Telescopes 7*. 2023. [2 citations]
10. Johnson A.B., **Thompson W**., Caputa K., et al.. "Exoplanet Detection and Characterization: Development and Results of a New Generation Imaging Fourier Transform Spectrometer." *Proceedings of Adaptive Optics for Extremely Large Telescopes 7*. 2023.
11. Li D., **Thompson W**., Savransky D., Marois C.. "Focal plane wavefront control for the Gemini Planet Imager 2.0 calibration system (CAL2)." *Proceedings of the SPIE*. 2023.
12. **Thompson W**., Marois C., Singh G., Lardière O., Gerard B., Fu Q., Heidrich W.. "Performance of the Fast Atmospheric Self Coherent camera at the NEW-EARTH lab." *Adaptive Optics Systems VIII 12185, 724-735*. 2022. [16 citations]

13. Singh G., **Thompson W**., Lardière O., et al.. “Pupil-plane LLOWFS simulation and laboratory results from NEW-EARTH’s high-contrast imaging testbed.” *Adaptive Optics Systems VIII 12185, 1592-1603*. 2022. [9 citations]
14. Marois C., Lardière O., **Thompson W**., Singh G., et al.. “Deployment of focal plane WFS technologies on 8-m telescopes.” *Adaptive Optics Systems VIII 12185, 594-603*. 2022. [21 citations]
15. Lardière O., Marois C., **Thompson W**., Singh G., et al.. “Optical design of SPIDERS, a Subaru Pathfinder Instrument.” *Adaptive Optics Systems VIII 12185, 1252-1262*. 2022. [10 citations]
16. Johnson A.B., Marois C., Gamroth D., Fitzsimmons J., Lardière O., **Thompson W**., et al.. “Blinking the fringes: initial development and results of the Ultra-Low Speed Optical Chopper.” *Adaptive Optics Systems VIII 12185, 2232-2239*. 2022. [9 citations]
17. Marois C., Gerard B., Lardière O., et al. (incl. **Thompson W**.). “Upgrading the Gemini Planet Imager calibration unit with a photon counting focal plane wavefront sensor.” *Adaptive Optics Systems VII 11448, 1366-1376*. 2020. [17 citations]
18. Lardière O., Gerard B., **Thompson W**., Marois C., et al.. “Optical design and preliminary results of NEW EARTH, first Canadian high-contrast imaging laboratory.” *Adaptive Optics Systems VII 11448, 1334-1341*. 2020. [11 citations]
19. **Thompson W**., Giordano M.. “AstroImages.jl.” *Proceedings of JuliaCon*. 2022.
20. **Thompson W**., Marois C.. “Extremely bright orbital guide beacons for extremely large telescopes.” *Adaptive Optics for Extremely Large Telescopes 6*. 2019. [5 citations]

White Papers

1. Marois C., Gerard B., **Thompson W**., Dong R., Metchev S., et al.. “Exoplanet Imaging: a technological and scientific road-map for finding Life signatures on other Worlds.” *Canadian Long Range Plan For Astronomy and Astrophysics White Papers*. 2019.

Presentations

Invited Talks: 6 · **Contributed Talks: 7** · **Posters: 13**

Invited Talks

- **2025**: “Plaskett Award Lecture” — Canadian Astronomical Society Annual General Meeting
- **2024**: “Orbit modelling to detect exoplanets” — University of Washington
- **2024**: “Orbit modelling to detect exoplanets” — University of British Columbia
- **2024**: “Orbit modelling to detect exoplanets” — Dominion Astrophysical Observatory Colloquium
- **2023**: “Review on spatial modulation techniques for coherent differential imaging” — Use of light coherence for high-contrast exoplanet imaging with ground-based telescopes (Paris)
- **2022**: “Deployment of focal plane wavefront sensing on 8-meter telescopes and beyond” — Wavefront Sensing in the VLT/ELT era VII (Porto)

Contributed Talks

- **2024**: “Real-time adaptive optics control with a high level programming language” — SPIE (Yokohama)
- **2024**: “Deploying focal plane wavefront sensing and coherent imaging at Subaru with SPIDERS” — Canadian Astronomical Society Annual General Meeting (Toronto)
- **2023**: “Deep orbital search for additional planets in the HR 8799 system” — AAS 241 (Seattle)
- **2022**: “Performance of the FAST Self Coherent Camera at the NEW-EARTH Lab” — SPIE Astronomical Telescopes + Instrumentation (Montreal)
- **2022**: “AstroImages.jl” — JuliaCon
- **2022**: “Julia for Adaptive Optics” — Julia in Astronomy & Astrophysics Research Symposium at JuliaCon
- **2021**: “Characterizing Exoplanets with an Imaging Fourier Transform Spectrograph” — Canadian Astronomical Society Annual General Meeting

Poster Presentations

- **2024**: “SPIDERS: A pathfinder 4th generation planet imager” — SPIE (Yokohama)
- **2024**: “Joint modelling to discover and characterise exoplanets: 51 Eri and Eps Eri” — Canadian Astronomical Society Annual General Meeting (Toronto)
- **2023**: “High-Spectral Resolution Dark Holes: Concept, Results, and Promise” — Adaptive Optics for Extremely Large Telescopes 7 (Avignon)
- **2023**: “Octofitter: Uniting all exoplanet detection methods under a single code” — Canadian Astronomical Society Annual General Meeting (Penticton)
- **2022**: “Limits on additional planets in HR8799” — In the Spirit of Lyot (Leiden)
- **2022**: “Combining Direct Imaging with orbital motion, radial velocity, and GAIA” — Canadian Astronomical Society Annual General Meeting (online)
- **2022**: “Imaging Planets with Kepler’s Laws” — NRC Herzberg Jamboree (online)
- **2021**: “Soft Realtime Control of an Exoplanet Imaging Instrument” — JuliaCon (online)
- **2020**: “Searching for Additional Outer Planets Around HR8799” — Canadian Astronomical Society Annual General Meeting (online)
- **2019**: “An EMCCD for Wavefront Sensing at 3kfps” — NTCO AGM (Penticton)
- **2019**: “Searching the HR8799 system for additional planets” — In the Spirit of Lyot (Tokyo)
- **2019**: “Is There Another Planet Hiding Around HR8799?” — CASCA (Montreal)
- **2019**: “Extremely Bright Orbital Guide Beacons for Extremely Large Telescopes” — AO4ELT6 (Quebec City)

Observing Programs

Instrument	Time	Program	Role
JWST/NIRCam	8.5 hours	Direct imaging and spectrum a true Jupiter-Saturn	Co-PI
JWST/NIRSpec, MIRI, NIRCam	26 hours		
JWST/NIRSpec	19.28 hours	Direct imaging and spectrum a true Jupiter-Saturn	Co-I
VLTI/GRAVITY	2 nights $\times$ 4 telescopes	ExoGRAVITY survey and follow-up of HR8799 f	
Keck II/NIRC2	3 half nights	AF Lep b M-band imaging, continued	Co-I
Keck II/NIRC2	4 half nights	AB Aur M-band imaging, continued	Co-I
Keck II/NIRC2	1 full night	Subaru/Keck Accelerating Stars Survey	
Keck II/NIRC2	2 half nights	AB Aur M-band imaging	Co-I
Keck II/NIRC2	3 half nights	AF Lep b M-band imaging	Co-I
Keck II/NIRC2	9 quarter nights	HR8799 planet search	Co-I
Keck II/NIRC2	2 half nights	Astrometric Monitoring of Imaged Planets	Co-I
Gemini Planet Imager	2 nights	Gemini Planet Imager Exoplanet Survey	

Mentoring

- **2026**: Zachary Oberhamer (masters)
- **2026**: Kaitlyn Hessel (phd)
- **2026**: Andre Fogal (phd)
- **2024**: Kaitlyn Hessel (masters)
- **2024**: Andre Fogal (masters)
- **2023**: Angus Bews (undergraduate)
- **2022**: Dori Blakely (phd)
- **2022**: Jensen Lawrence (undergraduate)

Teaching

- **PHYS248: Computer Assisted Mathematics and Physics** — University of Victoria (2023, 2022, 2019)
- **ASTR101: Introduction to the Night Sky** — University of Victoria (2020, 2019)

Service and Outreach

- **2023-present**: Colloquium committee member (2023-present) and chair (2025-present), NRC Herzberg, Dominion Astrophysical Observatory
- **2024**: Paper reviewed for Journal of Open-Source Software
- **2024**: Papers reviewed for the Astronomical Journal
- **2025**: Paper reviewed for Nature Astronomy
- **2025**: Paper reviewed for the Journal of Astronomical Telescopes, Instruments, and Systems (JATIS)
- **2024**: “Introduction to Julia for astronomy“ Software Workshop at University of Victoria
- **2023**: Public outreach talk: “Seeing is believing: capturing images of nearby solar systems“ at Royal Astronomical Society Vancouver Island Summer Star Party
- **2023**: co-host of “Introduction to Julia Workshop“ at AAS (Seattle)
- **2019-2022**: CASCA Graduate Student Committee representative for the University of Victoria
- **2020-2022**: Mentor to incoming graduate students
- **2022**: Created orbit animations for public outreach with more than 1M online impressions
- **2022**: Creator and maintainer of the @JuliaAstro official Twitter account
- **2021**: “Introduction to Julia for astronomy“ Software Workshop at University of Victoria
- **2019**: Judge, Vancouver Island Regional Science Fair
- **2019**: Volunteer for International Astronomy Day, UVic Observatory / Royal BC Museum

Software

- **Octofitter.jl**: Radial velocity, astrometry, proper motion, and direct imaging modelling software https://github.com/sefffal/Octofitte
- **PairPlots.jl**: Statistical visualization package https://github.com/sefffal/PairPlots.jl
- **PlanetOrbits.jl**: Orbital modelling and visualization package https://github.com/sefffal/PlanetOrbits.jl
- **AstroImages.jl**: Image processing, visualization, FITS, and astronomical coordinate handling package https://github.com/JuliaAstro
- **SNAP**: Signal to Noise Analysis Pipeline for reducing direct imaging data
- **Julia in Astronomy Tutorial Series**: Official Julia in Astronomy Tutorial Series

Previous Employment

Intern (ATD) — NRC Herzberg Astronomy and Astrophysics 2020

Developed control software for the NEW-EARTH Lab VIPER bench and demonstrated closed loop control of the self-coherent camera. **NSERC CREATE Intern — Nüvü Cameras** 2019

Contributed to development of a high speed, electron multiplying CCD camera (EMCCD) for adaptive optics applications. Deployed in GPI 2.0. **Submarine Network Platform Solutions — Ciena** 2017-2018

Lead development of software for characterizing submarine fiber optic cables and optimization of optical modem performance. **Summer Student — Queen’s University Department of Physics** 2017

Education research and lab development for undergraduate Physics labs. **Submarine Optical Systems R&D — Ciena** 2016

Development of software for optical spectrum analysis, instrument control, fiber link characterization, and signal optimization. **Submarine Optical Systems R&D — Ciena** 2015

Spectral analysis, instrument control, dispersion compensation, and optimization of optical launch power.

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